

Global Facilities - Design & Construction Standard - CSA

- Structural Steel Works Specifications

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[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

- 1.1 This document provides a Required Standard and Standard Guideline on Material and Design Specification that shall be followed and complied in the Design, Engineering and Construction for Structural Steel Works as part of any green and/or brown field construction projects contract requirement.
- 1.2 The standard is applicable for any green and/or brown field construction projects as per required in the contract.
- 1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings and Specification provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.
- 1.4 The established Specifications Standard for each locale shall also comply with Micron's in-house:
 - 1.4.1 RBA Code of Conduct requirements for Semiconductor Manufacturing
 - 1.4.2 Environmental, Health & Safety Standards
 - 1.4.3 Operational Requirements /SOP (Standard Operation Procedures)
 - 1.4.4 Security Protocol
- 1.5 The Specifications Standard to be adopted in each local shall not violate any local codes or by-laws and reflects the current semi-conductor industry's best practices.

- 1.6 Where there is the option to adopt either the MasterSpec or in-region specifications standard, the AE (Architectural & Engineering Consultants) shall propose the most suitable solution based on safety, quality, time & cost to Micron's Designated Project Team (DPT) for review and acceptance. Otherwise, the more stringent option shall apply.

2.0 Scope

- 2.1 This document for structural steel works specification covers all necessary related design, engineering, procurement, construction, installation and completion of piling, substructure, superstructure, and roof structure to full compliance of the contract requirement and required for the project. It extended to cover any green and/or brown field construction projects.
- 2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

- 3.1 Definitions of acronyms used in this document are listed as follows:

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
AE	External Architectural & Engineering Consultants
DPT	Micron's Designated Project Team
GFCT	Micron's Global Facilities Construction Team
EHS	Micron's Environmental Health & Safety Team
QP/PE	External Qualified Person (professional architect/engineer)
SME	Micron's Subject Matter Expert
TM	Micron's Team Members

4.0 Codes and Standards

All Codes and Standards referred to shall be deemed to be the latest edition at the time of tender.

The design and erection of structural steelworks shall comply with the requirements of BS: 1, SS EN 1993, SS EN 1994 and BS EN 1090-2 or equivalent local/international code requirements.

5.0 Materials

- 5.1 Structural steelwork shall comply with the appropriate Standards shown in **Table 1** below:

Table 1

Form	Material Quality	Dimensions	Dimension Tolerances
I & H	BS EN 10025	BS EN 10365	BS EN 10034
Channel		BS EN 10365	BS EN 10279
Angles		BS EN 10056-1	BS EN 10056-2
Plate & Flats		Not applicable	BS EN 10029
Structural Hollow Sections (Hot	BS EN 10210-1	BS EN 10210-2	BS EN 10210-2

- 5.2 All steelwork shall be fabricated from new sections.
- 5.3 The Builders shall submit to the Engineer certified copies of Mill certificates before commencement of fabrication works.
- 5.4 Structural Fasteners
- Bolts shall be Grade 8.8 or Grade 10.9 high tensile bolts as specified. These shall comply with BS 3692. The mechanical properties of the bolts shall comply with BS 3692 and BS EN ISO 898-1.
 - Nuts shall be Grade 8 or Grade 12, complying with BS 3692
 - Plain and tapered steel washers shall comply with BS 4320
 - Spring washers shall comply with BS 4464
 - High Strength Friction Grip (HSFG) bolts and accessories shall comply with BS EN 14399.
 - Holding down bolts shall comply with BS 7419.
- 5.5 Electrodes
- Electrodes for metal arc welding shall comply with BS EN ISO 2560.

6.0 Submittals

- 6.1 The Builder shall prepare and submit to the Engineer for approval the following at least 4 weeks before the commencement of fabrication of steelwork :
- Shop drawings showing details of the works.

- b. Fabrication method statements and welding procedures.
 - c. Cranage details
 - d. Calculations on handling and erection stresses.
 - e. Temporary steelworks/towers, guys and bracings proposed for use during erection.
- 6.2 The Builder's Professional Engineer shall endorse all relevant drawings, calculations, and submittals.
- 6.3 Approval of the Builder's drawings shall not in any way relieve the Builder of his responsibility for any errors, which his drawings or calculations may contain.

7.0 Fabrication

- 7.1 Steel fabricators shall be accredited as at least Class 2 Fabricators under the Structural Steel Fabricators Accreditation Scheme of the Singapore Structural Steel Society and BCA for fabrication works.
- 7.2 Steel fabrication and accuracy of fabrication shall conform to BS EN 1090-2.
- 7.3 Joints shall be prepared in accordance with BS EN ISO 9692.
- 7.4 The Builder shall obtain the permission of the Engineer for commencement of fabrication. Such permission is normally not given before the shop drawings are approved.
- 7.5 All steelwork shall be fabricated to accuracy within the allowable tolerances so as to ensure the absence of permanent erection stresses in the structures.
- 7.6 Details for ensuring the soundness and cleanliness of the pile base prior to concreting;
- The Engineer's permission shall be obtained if manual thermal cutting is to be carried out. Manual thermal cutting is permitted only for wall ends of beams, notching and such other application as may be approved.
- For edges to be welded, manual thermal cutting is not permitted.
- Edges of plates cut by flame to be welded shall be machined prior to welding. The amount of metal to be removed shall be subject to the Engineer's approval.
- Process of thermal cutting for Grade S 355 steel shall be controlled to prevent excessive hardening of the edges. The Vickers hardness value shall not exceed 350 when tested to BS EN ISO 6507 with 10 kgf
- 7.7 Faying surfaces of HSFG joints in steel 25mm thick or more shall be checked for deformities such as bowing twist or rippling. Any deformed surfaces shall be machined flat to achieve the design value of slip factor.
- 7.8 Ends of compression members at splices, joints, caps and bases shall be right angles to the axis of the member and shall have butting faces in contact except for small areas with clearances not exceeding 0.15mm.
- 7.9 Stiffener members with ends abutting flanges shall be cut and ground to ensure a tight fit at the contacting surfaces.

- 7.10 All burns and sharp arises shall be removed by grinding.
- 7.11 For trusses and lattice girders, which are to be assembled before delivery to site, the Builder shall check the fit, profile and camber before and after the making of permanent connections.
- 7.12 The Builder shall give a minimum of 2 days' notice to the Engineer for inspection of the steelworks prior to dispatch. The Builder shall afford the Engineer or his representatives the right to visit the Builder's or his supplier's factory for the purposes of inspection.
- 7.13 To facilitate erection and inspection of steelworks, all steel members shall be clearly marked in accordance with marking drawings prepared by the Builder.
- 7.14 All steelworks shall be trial-assembled before erection.
- 7.15 Acceptance by the Engineer does not relieve the Builder of his obligation to replace or rectify defective material or poor workmanship nor does it absolve the Builder from errors in dimensions and sizes of finished materials.

8.0 Welding

- 8.1 All surfaces to be welded shall be free of rust, grease and dirt.
- 8.2 Metal arc welding of steel shall comply with the requirements of BS EN 1011-1 and 1011-2.
- 8.3 Approved written Welding Procedures shall be available in accordance with BS EN ISO 15607.
- 8.4 For types of welding not covered by a Eurocodes and Singapore Standards, the Builder shall:
 - a) Follow as closely as possible the procedure for metal arc welding specified in the Eurocodes and Singapore Standards.
 - b) Propose for the Engineer's approval, the type method and procedure of welding.
 - c) Conduct destructive testing of fusion welded joints to BS EN ISO 4136, BS EN ISO 5173, BS EN ISO 5178, BS EN ISO 9015, BS EN ISO 9016, BS EN ISO 9017, BS EN ISO 9018 and BS EN ISO 17639 as directed by the Engineer.
- 8.5 All welding shall be done in a manner such that negligible distortion is produced. Corrections if required shall be at the Builder's expense.
- 8.6 Weld deposits shall have mechanical properties not less than the minimum specified for the parent metal.
- 8.7 Electrodes for manual arc welding of steel shall comply with the requirements of BS EN ISO 2560.
- 8.8 All electrodes shall be used in strict accordance with the electrode manufacturer's recommendations.
- 8.9 No welding to members shall be carried out unless approved or shown on drawings. Approved temporary welded connections shall be removed

without damage to the main metal.

- 8.10 The length of tack welds shall be 50 mm or four times the thickness of the thicker metal, whichever is the larger.
- 8.11 Butt welds shall be made using run-on and run-off plates to ensure full throat thickness at the ends. Such plates shall be of the same material as the steel being welded and prepared in the same manner as the parts being joined. After completion of the welding and removal of the plates, the surfaces of joined parts shall be ground smooth.
- 8.12 All weld slags shall be removed by light hammering and/or wire brushing ensuring that the welds are not disturbed.
- 8.13 Welding shall be carried out by skilled and qualified welders who had been tested to BS EN ISO 9606-1 or BS 4872-1 as appropriate. Evidence of qualification shall be submitted prior to the commencement of works.
- 8.14 Only qualified welders tested by approved accredited laboratories (e.g. SETSCO or Singapore Test Services Pte Ltd) shall perform welding.
- 8.15 The Builder shall appoint an approved independent Accredited Inspection and Testing Agency (e.g. SETSCO) to inspect and test the quality of all welded connections and joints.

9.0 Ordinary Bolted Assemblies

- 9.1 All bolted connections shall be made in accordance with SS EN 1993 unless otherwise specified.
- 9.2 The strength grade combination of bolts/nuts/washers for any connection shall comply with the requirements or recommendations of the relevant British Standards.
- 9.3 The bolt length shall be chosen such that, after tightening, at least one thread plus the thread run-out will be clear between the nut and the unthreaded shank of the bolt and at least one clear thread shall show above the nut.
- 9.4 When bolts are pre-set into concrete element, provision shall be made to the normal protrusion needed to accommodate the nuts, washers plus a further tolerance for the concrete elements.
- 9.5 Nut or bolt head shall be provided with a minimum of one washer placed under the part being rotated.
- 9.6 For surfaces sloping 3° or more from a plane at right angles to the bolt axis, tapered or suitably shaped washers shall be used to achieve satisfactory bearing.
- 9.7 All bolt holes must be drilled, or sub-punched and reamed.
- Holes in materials not exceeding 6mm thickness may be punched without reaming.
- Finished holes shall be free of ragged edges or burrs and shall not exceed 2mm larger than the nominal diameter of the bolt.

10.0 High Strength Friction Grip Assemblies

- 10.1 High Strength Friction Grip Bolts shall conform to BS EN 14399, as appropriate. HSFB bolts shall be tightened using methods approved by the Engineer.
- 10.2 The use of friction grip bolts shall comply with BS EN 1993-1-8.
Hardened plain washer shall be placed under part being turned.
Tightening may be by part-turn method, or with load indicating devices used in accordance with the manufacturer's recommendations.
Load indicator shall be placed under the head of the bolt and the nut is rotated on a hardened steel washer.

11.0 Shear Studs

- 11.1 Diameter 19mm studs used in composite construction shall be the headed type with the following properties after being formed:
 - (a) Minimum yield strength : 350 N/mm².
 - (b) Minimum ultimate tensile strength : 450 N/mm².
 - (c) Elongation of 15%.

12.0 Erection

- 12.1 Accuracy of erected steelwork shall be in accordance with BS EN 1090-2.
- 12.2 The erection of the steelwork shall be planned and carried out so as to ensure safe- working conditions at all times. The Builder shall comply with local requirements of Workplace Safety and Health Act and Factories Act.
- 12.3 Permission shall be obtained from the Engineer for commencement of erection at site.
- 12.4 Prior to erection, the Builder shall ensure that all works by others to which the steelworks is placed, connected or abutted are correct in lines and level. Any discrepancy shall be reported to the Engineer immediately.
- 12.5 Erection shall be carried out expeditiously by a competent crew with adequate crane and equipment.
- 12.6 Steelworks shall not be distorted during handling and erection. Stress limits specified in SS EN 1993 shall not be exceeded.
- 12.7 Braces and guys shall be provided wherever necessary to ensure stability and safety at all times.
- 12.8 Permanent bolting shall not be carried out until all members concerned have been properly aligned and in their final positions.
- 12.9 Any individual site connections made more than one stage during erection shall be approved by the Engineer.
- 12.10 Columns shall be plumbed using sawn steel packs and wedges not larger

than necessary.

The grouting space below the base plate shall not be less than 25mm and not greater than 50mm.

- 12.11 Ends of hold-down bolts/anchors shall protrude a minimum of 25mm above the nuts.
- 12.12 All loose and foreign materials shall be removed from the hold-down bolts, base plates and top of column stumps before grouting commences.
- 12.13 Column bases shall be grouted with approved non-shrink grout. Preparation, mixing and application procedures shall be in accordance with the recommendations of the manufacturer.
- 12.14 The Engineer may order adjustments to compensate for dimensional discrepancies caused by temperature variations during erection.

13.0 Test

- 13.1 The Builder shall include in his price for the costs of material and mechanical tests including provision of facilities, attendance etc. for inspection and testing, as required by the Engineer.
- 13.2 All tests shall be carried out by accredited testing agencies approved by the Engineer. Tests results shall be submitted to the Engineer immediately.
The Builder shall appoint at his own cost, an Independent Inspection and Testing Agency to inspect, check and test the quality of the steelworks including welded joints and bolted connections.
- 13.3 The Builder shall carry out some or all of the following tests to Eurocodes and Singapore Standards recommendations, as directed by the Engineer:
 - a. Chemical Analysis
 - b. Tensile Stress
 - c. Impact Tests
 - d. Visual Inspection of All Welds
 - e. Non-destructive Testing of Welds
- 13.4 If a sample fails a specified test, the Engineer may at his discretion, reject in whole or part the consignment which the sample represents.
- 13.5 The Builder shall carry out non-destructive tests in the presence of the Engineer and / or his Representative.

14.0 Schedule of Construction Tests

- 14.1 The Builder shall allow and conduct all construction tests as prescribed in the Building Control Regulations and the schedule **Table 2** below:

Table 2

Tests to be carried out	Standards to be used	Minimum frequency of testing / the numbers of testing required
Structural Steel: a) Chemical analysis b) Tensile test	BS EN 10025, for hot rolled sections BS EN 10210-1, for hot finished structural hollow sections	1 sample per section type per source
Weld Quality:	BS EN ISO 5817 Quality Level B	
a) Visual Inspection (By approved qualified welding Inspector)	BS EN ISO 17637 for compliance with BS EN ISO 5817	100%
Non-destructive testing of welds	BS EN ISO 17635	
a) MPI/DPI – For fillet welds	MPI – BS EN ISO 17638 & BS EN ISO 23278 or DPI – BS EN ISO 3452 & BS EN ISO 23277	10% (When a crack is found, 100% MPI will require)
b) Ultrasonic Exam. – For fillet welds	BS EN ISO 17640 & BS EN ISO 11666	5 %
c) Radiography Exam. – For Butt weld	BS EN ISO 17636 & BS EN ISO 10675	100 %
Fastener (Bolt) :		
a) Chemical analysis		1 sample per size per source

Tests to be carried out	Standards to be used	Minimum frequency of testing / the numbers of testing required
b) Tensile test	BS EN ISO 898-1 (BS 3692)	3 samples per size per consignment per source

15.0 Protective Treatment

15.1 Performance requirements for protection of steelwork.

Durability range – minimum 10 years

Atmospheric environmental class - Category C (medium) under EN ISO 12944.

15.2 Protective works shall be carried out in accordance with BS EN ISO 14713 and/or BS EN ISO 12944 as appropriate.

15.3 Surface preparation Grade shall be as specified in the drawing.

15.4 Unless otherwise specified, all surface preparation of members/parts shall be Grade Sa 2½, very thorough abrasive blast cleaning in accordance with BS 7079.

15.5 Surface preparation Grade St 3, very thorough hand and power tool cleaning of members/parts shall be in accordance with BS 7079.

15.6 Prefabrication paint coating is required, and shall fully be compatible with protection system.

16.0 Free-Proof Coating

Sprayed-on fire-proofing, unless approved by the Engineer, shall not be used.

17.0 Document Control

17.1 Approval: GLOBAL_FAC_GFTT_APPR

17.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCGF11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

17.3 Notification:

All Micron Team Members at all Sites

17.4 Retention

Adherence to the requirements specified in

[Global Facilities - Document Control Procedures](#)

17.5 Review

Biennially (two years)

18.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New Document	07/06/2022